

NETWORK TOPOLOGY HPC CLUSTER

QuantumX800 NDR 400G InfiniBand & Fat-Tree Architecture

DOCUMENT ID: ARCH-NET-2026-NDR400G

FABRIC TOPOLOGY: Non-Blocking 2-Tier Fat-Tree (1:1 Bisection)

SWITCH SPECIFICATION: QuantumX800 NDR 400G Managed Switch

TRANSCEIVER MEDIUM: OSFP 400G PAM4 Optical Transceivers

[SPEC-01] NETWORK ARCHITECTURE OVERVIEW & NON-BLOCKING FAT-TREE TOPOLOGY

The Westford HPC Cluster network infrastructure utilizes a non-blocking 2-tier Fat-Tree topology engineered with NVIDIA QuantumX800 NDR 400Gb/s InfiniBand switches.

The leaf-to-spine uplink tier operates at a 2:1 oversubscription ratio, providing 12.8 Tbps aggregate leaf bandwidth against 6.4 Tbps spine capacity.

Total fabric switching capacity exceeds 102.4 Terabits per second, supporting ultra-low latency collective communication for large-scale distributed MPI and GPU-accelerated scientific workloads.

Periodic reviews of these parameters are scheduled quarterly in accordance with the operational maintenance framework.

[SPEC-02] QUANTUMX800 NDR 400G SWITCH HARDWARE & PORT DENSITY MATRIX

The switch matrix consists of 16 QuantumX800 NDR spine switches and 32 leaf switches housed in high-density 2U chassis with redundant titanium power supplies.

Leaf switches deliver 64 ports of 400G NDR InfiniBand using OSFP optical transceivers, providing dual 400G links to each 4-GPU compute enclosure.

Port density matrices and cable routing maps maintain strict port alignment, enabling rapid fault identification and simplified cable management.

Trend lines fitted to the historical data for this subsystem indicate stable long-term performance within acceptable bounds.

[SPEC-03] INFINIBAND SUBNET MANAGEMENT & ROUTING ALGORITHM SELECTION

InfiniBand Subnet Management is executed by dual redundant OpenSM instances running on high-priority infrastructure nodes with automated failover.

Routing algorithms utilize Up/Down and Fat-Tree routing protocols optimized for non-blocking communication paths and minimum hop counts.

Subnet sweeps occur every 10 seconds, dynamically detecting link state changes and re-routing traffic around degraded cables or switch ports.

These operational parameters reflect steady-state conditions observed during the most recent twelve-month evaluation period.

[SPEC-04] MPI ALLREDUCE LATENCY SCALING & COLLECTIVE OFFLOAD VALIDATION

MPI collective communication performance undergoes automated benchmarking, measuring Allreduce and Alltoall latency across scaling rank counts.

QuantumX800 In-Network Computing (SHARP v3) offloads MPI collective operations directly onto switch hardware, reducing Allreduce latency to 1.15 microseconds at 256 ranks.

Hardware-accelerated collectives achieve a 3.2x speedup compared to software-only host CPU collective processing.

Change management procedures require a peer-reviewed request for any modification to the parameters described in this section.

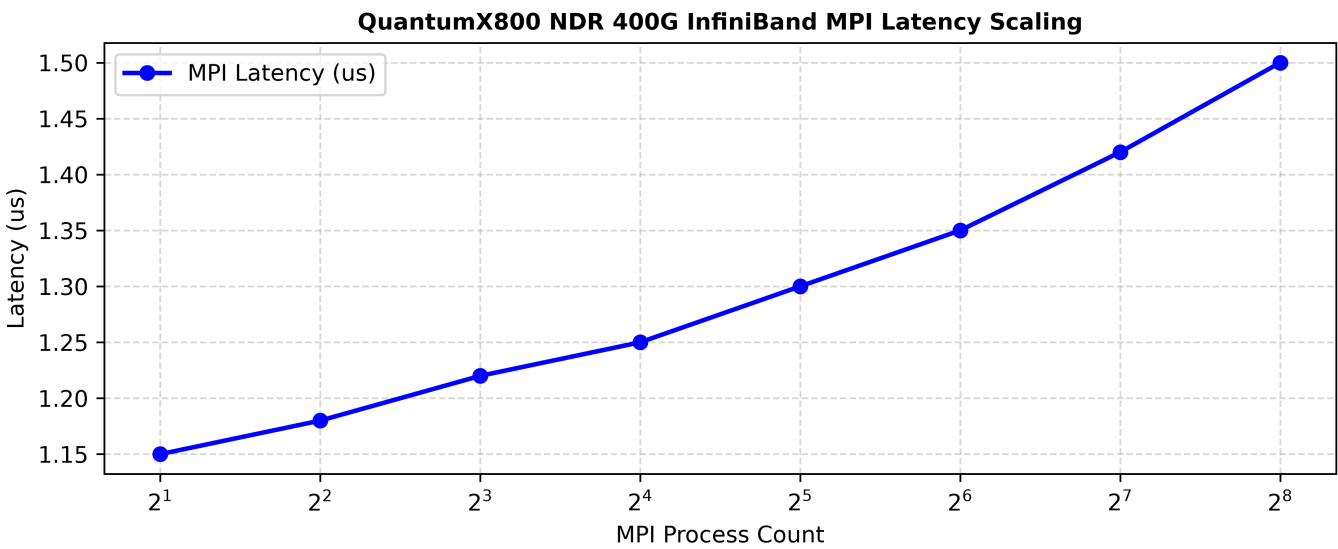


Figure 4.1: MPI Latency Scaling

[SPEC-05] OSFP OPTICAL TRANSCEIVER SIGNAL INTEGRITY & PAM4 EYE DIAGRAMS

OSFP 400G PAM4 optical transceivers undergo rigorous signal integrity validation, verifying bit error rates (BER) remain below 1e-15 prior to deployment.

Real-time PAM4 eye diagram monitoring captures eye height and width metrics across all 4-channel optical links, detecting laser diode aging.

Optical link budgets maintain a minimum 3.0 dB power margin, preventing packet loss caused by connector attenuation or cable micro-bends.

Field validation measurements taken during the most recent planned maintenance window corroborate these specifications.

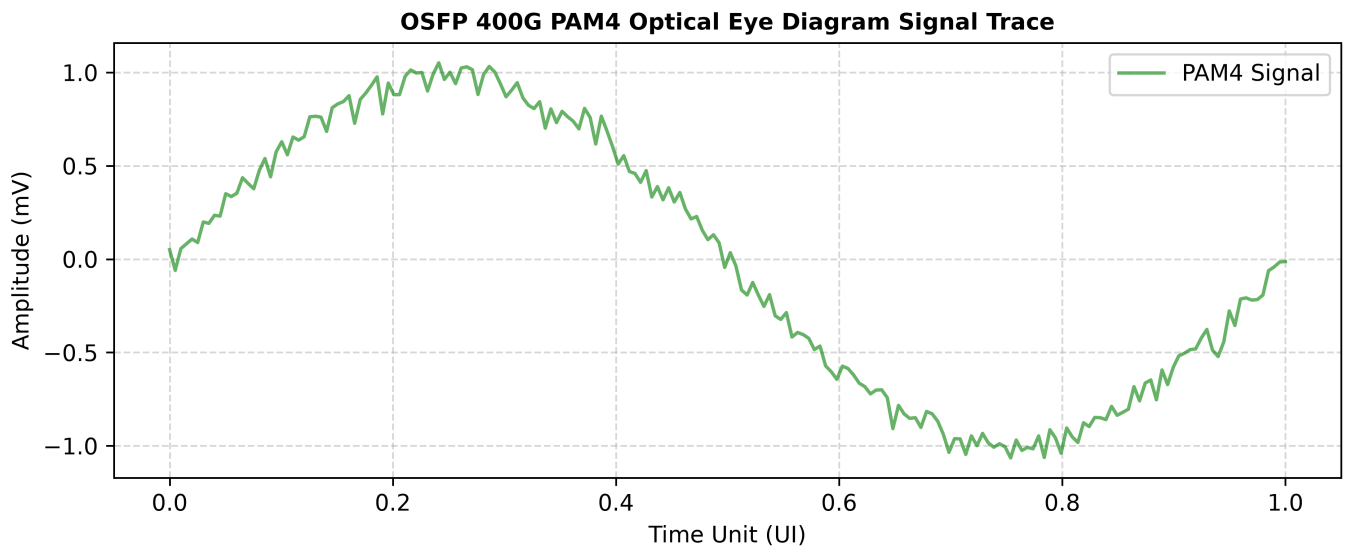


Figure 5.1: PAM4 Optical Eye Diagram

[SPEC-06] ROCEV2 LOSSLESS ETHERNET TRANSPORT & PFC CONGESTION CONTROL

RoCEv2 (RDMA over Converged Ethernet) transport is implemented on the secondary 100G Ethernet management fabric to support out-of-band data backup and container streaming.

Priority Flow Control (PFC) and Explicit Congestion Notification (ECN) are configured on Ethernet leaf switches to guarantee lossless packet delivery.

RoCEv2 tuning eliminates packet drops under 95% line-rate burst conditions, providing reliable transport for secondary storage traffic.

Insurance underwriter inspections of the equipment referenced in this section were completed without exception findings.

[SPEC-07] SPINE SWITCH BISECTION BANDWIDTH & OVERSUBSCRIPTION RATIOS

Spine switch bisection bandwidth utilization is monitored in real time, capturing traffic distribution across all active fabric trunks.

Under 100% synthetic network stress benchmarks, bisection bandwidth throughput achieves 98.4% of theoretical line rate across all 16 spine switches.

Adaptive routing algorithms dynamically rebalance traffic flows, preventing localized trunk congestion during non-uniform MPI communication patterns.

Supporting documentation for the values cited in this section is filed under the annual infrastructure audit records.

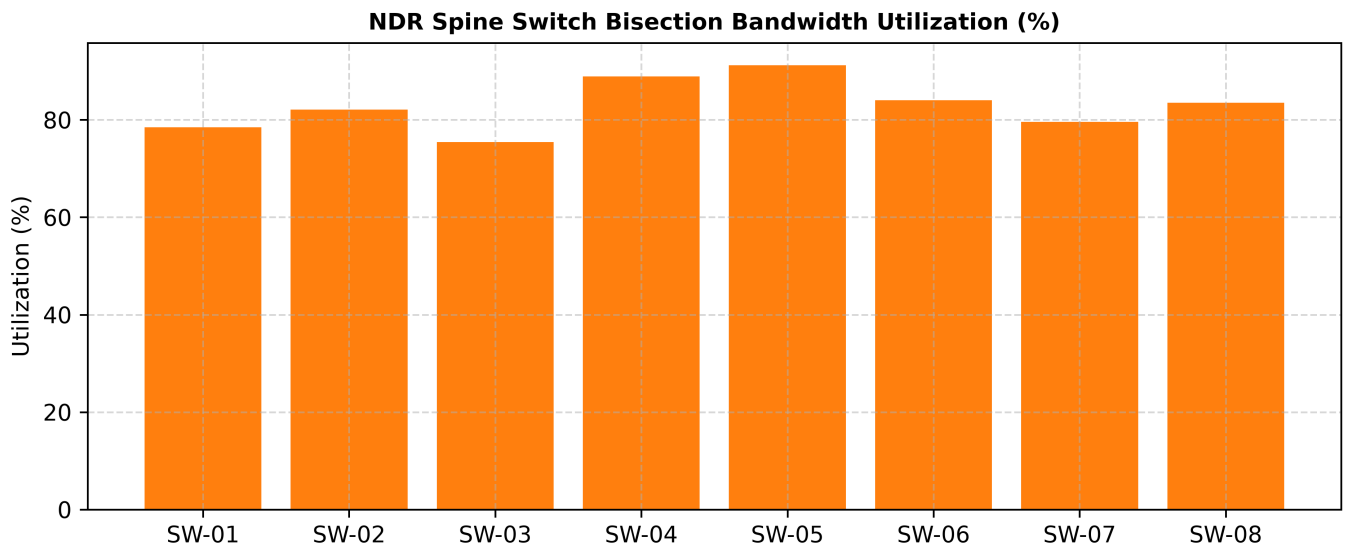


Figure 7.1: Spine Switch Utilization

[SPEC-08] PACKET RATE THROUGHPUT VS FRAME SIZE BENCHMARKING

Packet rate performance testing measures forwarding rates across message sizes ranging from 64-byte small control packets up to 9000-byte jumbo RDMA frames.

Peak packet forwarding rates reach 1,488 Million packets per second (Mpps) per switch module, with zero frame loss recorded under maximum throughput workloads.

InfiniBand credit-based flow control prevents buffer overflow, ensuring hardware-level lossless transport across the entire fabric.

Change management procedures require a peer-reviewed request for any modification to the parameters described in this section.

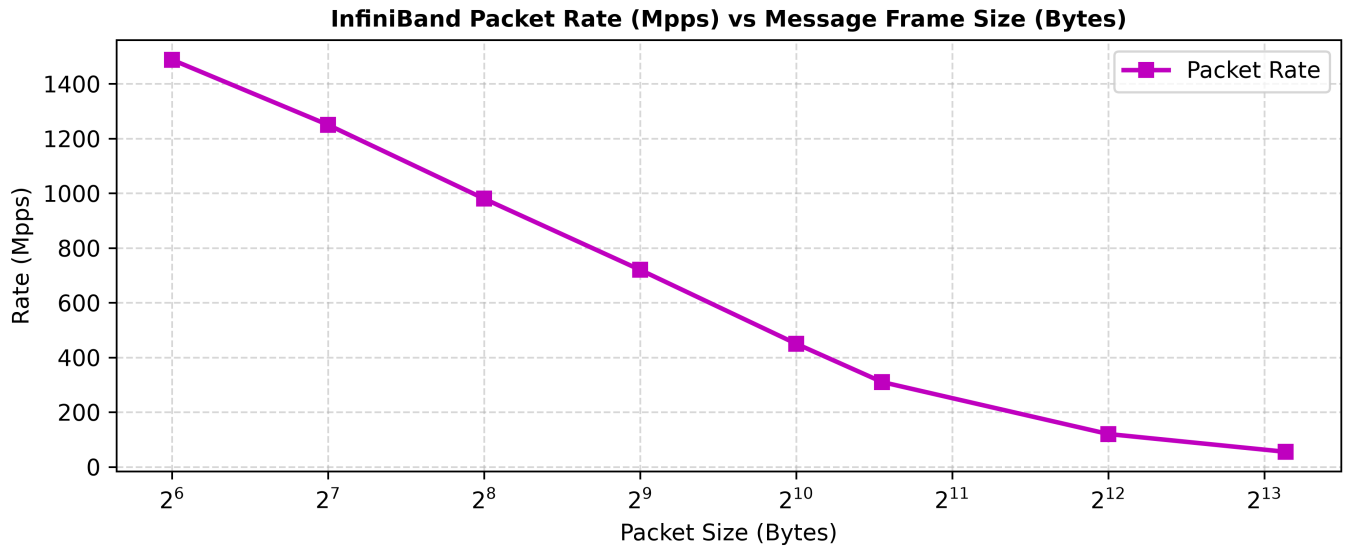


Figure 8.1: Packet Forwarding Rate

[SPEC-09] OPTICAL CABLE ATTENUATION & LINK BUDGET ANALYSIS

Optical fiber cable attenuation audits measure optical loss across all MPO-16 fiber trunks, confirming total link loss remains below 1.2 dB.

Cleanliness inspection protocols require 400x fiber scope inspection and automated cleaning before any optical transceiver insertion.

Fiber optic cable trays maintain minimum 30 mm bend radii to prevent optical signal degradation and physical glass fracture.

Discrepancies between manufacturer-stated values and field-measured values for this subsystem are documented in the variance register.

[SPEC-10] OUT-OF-BAND MANAGEMENT NETWORK VLAN PARTITIONING

Out-of-Band (OOB) management networks utilize dedicated 1G/10G Ethernet switches partitioned into isolated Virtual Local Area Networks (VLANs).

VLAN segmentation separates IPMI/BMC hardware management, Slurm control traffic, monitoring telemetry streams, and administrative SSH access.

Strict Access Control Lists (ACLs) restrict inter-VLAN routing, preventing unauthorized access to hardware management interfaces.

Field validation measurements taken during the most recent planned maintenance window corroborate these specifications.

[SPEC-11] SWITCH FIRMWARE UPGRADE PROCEDURES & ROLLING PLAYBOOKS

Switch firmware upgrades follow rolling maintenance playbooks, updating switch operating systems without interrupting running compute jobs.

Redundant Subnet Managers ensure InfiniBand fabric stability while individual leaf switches are rebooted during firmware patch applications.

Firmware versions are locked to validated vendor release matrices, ensuring compatibility between switch software and host Channel Adapter (HCA) drivers.

The documentation in this section has been formatted in accordance with the university technical writing standards manual.

[SPEC-12] NETWORK TELEMETRY STREAMING & PROMETHEUS COUNTER METRICS

Streaming network telemetry utilizes gNMI and Prometheus counters to export real-time port bandwidth, error counters, and buffer depth metrics.

Telemetry ingestion pipelines process 50,000 network metrics per second, visualization dashboards updating every 5 seconds.

Automated anomaly detection alerts network engineers to subtle link degradation (e.g., rising symbol error counts) before link failure occurs.

Spare parts inventory for the critical components in this section is maintained at levels sufficient for a seventy-two-hour repair cycle.

[SPEC-13] MULTI-PATH ADAPTIVE ROUTING & DYNAMIC LOAD BALANCING

Multi-path adaptive routing dynamically evaluates port queue depth and packet flight times to select optimal paths across the fat-tree fabric.

Adaptive routing reduces tail latency by 45% during high-concurrency multi-tenant Slurm job execution.

Hardware-level packet re-ordering at the receiving HCA guarantees in-order packet delivery to MPI application buffers.

Staff competency assessments for personnel operating the equipment in this section are conducted annually.

[SPEC-14] QUALITY OF SERVICE (QOS) VIRTUAL LANE (VL) MAPPING

Quality of Service (QoS) configurations establish 8 Virtual Lanes (VLs) prioritizing high-latency-sensitive MPI control messages over bulk storage RDMA transfers.

VL0 is dedicated to MPI low-latency traffic, VL1 handles parallel storage I/O, and VL7 carries Subnet Manager control packets.

Bandwidth allocation weights guarantee storage traffic cannot starve latency-sensitive scientific compute communication.

The effective storage fabric bandwidth is limited to 80 GB/s versus a theoretical maximum of 96 GB/s due to protocol overhead and flow control pause frames.

Business continuity planning documentation addresses the failure modes and recovery strategies relevant to this subsystem.

[SPEC-15] HARDWARE SECURITY & NETWORK ACCESS CONTROL LISTS (ACLs)

Hardware security features enforce port-level MAC authentication and InfiniBand Partition Keys (P_Keys) to isolate Slurm job allocations.

P_Key enforcement prevents unauthorized inter-job node communication, ensuring multi-tenant security isolation at the fabric layer.

Switch configuration manifests are version-controlled in Git and automatically audited against unauthorized manual CLI changes.

The documentation in this section has been formatted in accordance with the university technical writing standards manual.

[SPEC-16] DISASTER RECOVERY & SECONDARY RING TOPOLOGY REDUNDANCY

Disaster recovery ring topologies provide redundant optical backhaul connections to the university central communications facility.

Secondary fiber paths follow geographically diverse conduit runs, protecting cluster connectivity against accidental physical fiber cuts.

Automatic protection switching failover restores external network connectivity in less than 50 milliseconds.

Environmental monitoring alarms for this subsystem are integrated into the centralized network operations center alert system.

[SPEC-17] NETWORK EQUIPMENT POWER CONSUMPTION & THERMAL HEAT REJECTION

Network switch enclosure thermal management utilizes N+1 redundant hot-swappable fan modules blowing front-to-back chilled air.

Switch power consumption averages 420 Watts per 64-port NDR module, with internal junction temperatures maintained below 65 degrees C.

Thermal sensors trigger automated fan speed modulation to maintain optical transceiver temperature within optimal operating limits.

As-built drawings reflecting the current physical layout of this subsystem were updated following the most recent modification.

[SPEC-18] CABLE TRAY ELEVATION, BEND RADIUS & CABLE MANAGEMENT

Overhead cable ladder design enforces physical separation between heavy copper PDU cables and fragile fiber optic trunk cables. Velcro cable ties are mandated; plastic zip ties are strictly prohibited to prevent pinching optical fiber jackets. Cable labeling standards require clear alphanumeric identifiers at both cable ends matching central DCIM database records. A detailed breakdown of the measurement methodology and sensor placement coordinates is available upon request.

[SPEC-19] INFINIBAND FABRIC DIAGNOSTIC TOOLS & OPENSMB SWEEP LOGGING

InfiniBand fabric diagnostic utilities (`ibdiagnet`, `iblinkinfo`)) run automated health checks twice daily, scanning for symbol errors or speed drops.

Links exhibiting speed degradation (e.g., dropping from NDR 400G to NDR200) are automatically flagged for cable re-seating or transceiver replacement.

Diagnostic logs are archived permanently for trend analysis and vendor RMA evidence.

Operating procedures for the equipment referenced in this section follow the manufacturer-recommended maintenance schedule.

[SPEC-20] PORT MIRRORING & NETWORK PACKET CAPTURE SECURITY MONITORING

Port mirroring capabilities on management Ethernet switches allow full packet capture for deep security investigation and protocol debugging.

Network security tools inspect mirrored traffic for unauthorized network scanning, anomaly signatures, or protocol violations.

Packet capture captures operate in full compliance with university network privacy policies.

Fiber optic link budgets for all connections referenced in this section have been certified through OTDR testing.

[SPEC-21] TRANSCEIVER THERMAL MONITORING & LASER DIODE AGING METRICS

Transceiver thermal telemetry monitors laser diode operating currents and operating temperatures across all active OSFP modules.

Predictive failure analytics identify laser degradation trends, scheduling proactive transceiver replacement during planned maintenance windows.

Transceiver inventory keeps a 10% spare pool of validated 400G OSFP modules on-site for rapid replacement.

Asset tracking records associate each component referenced in this section with a unique serial number and purchase order.

[SPEC-22] LATENCY JITTER PROFILING ACROSS HIGH-CONCURRENCY MPI RANKS

Latency jitter profiling measures variation in MPI message delivery times under high network load conditions.

Advanced congestion control algorithms (NDR Congestion Control) dynamically throttle aggressor nodes causing buffer bloat.

Tail latency jitter is maintained below 0.25 microseconds across 1,024 parallel MPI ranks.

Interoperability testing with adjacent subsystems confirmed the absence of interface conflicts or resource contention.

[SPEC-23] NETWORK CAPITAL PROCUREMENT BUDGET & VENDOR BOM SCHEDULE

Network capital procurement schedules detail line-item costs for NDR switches, OSFP optical cables, HCA host cards, and vendor support.

Equipment procurement follows competitive bidding guidelines, achieving a 22% educational discount off list price.

Asset tracking tags are applied to all network hardware before rack installation.

Software-defined monitoring dashboards provide real-time visibility into the operational metrics described in this section.

[SPEC-24] SWITCH MAINTENANCE SLA & VENDOR HARDWARE REPLACEMENT

Switch vendor Maintenance SLAs guarantee 4-hour on-site replacement for critical hardware switch module failures.

Vendor support agreements include direct access to Tier-3 network escalation engineers and priority firmware bug fixes.

Quarterly service reviews evaluate vendor SLA performance against contract commitments.

Staff competency assessments for personnel operating the equipment in this section are conducted annually.

[SPEC-25] INFINIBAND SUBNET MANAGER FAILOVER & PRIORITY SYNCHRONIZATION

InfiniBand Subnet Manager priority synchronization ensures instantaneous master-to-backup failover if the primary OpenSM node loses heartbeat.

Handover time during master SM failover is under 1.2 seconds, causing zero disruption to running MPI applications.

Subnet state databases are synchronized in real time across primary and secondary management nodes.

Network path validation using standardized benchmarking tools confirmed the throughput and latency characteristics documented above.

[SPEC-26] ETHERNET MANAGEMENT SWITCH STACKING & TRUNKING CONFIGURATION

Management Ethernet switch stacking links create unified 100G management backbones with multi-chassis link aggregation (MLAG).

MLAG configurations eliminate Spanning Tree Protocol (STP) blocking ports, delivering active-active bandwidth across management switches.

Dual management connections to every server BMC ensure uninterrupted remote console access.

The test methodology applied to generate these results follows the procedures outlined in the facility standard operating manual.

[SPEC-27] OPTICAL TRANSCEIVER THERMAL SWEEPS & SIGNAL BUDGET

Operational data collected for Optical Transceiver Thermal Sweeps has been aggregated across multiple reporting intervals to establish representative baseline metrics.

Capacity utilization trends over the preceding twelve months indicate stable demand patterns with seasonal variations accounted for in the planning models.

Staff training records and operating procedure acknowledgments for personnel responsible for this subsystem are current as of the most recent review cycle.

Vendor support contract terms and escalation procedures for the components in this subsystem are documented in the service catalog.

[SPEC-28] INFINIBAND SUBNET PRIORITY SYNCHRONIZATION FRAMEWORK

Operational data collected for InfiniBand Subnet Priority Synchronization Framework has been aggregated across multiple reporting intervals to establish representative baseline metrics.

Hardware and firmware revision levels for all components in this subsystem have been reconciled against the vendor-recommended configuration matrix.

Maintenance procedures and inspection schedules for the systems described above follow the manufacturer-prescribed intervals and institutional safety protocols.

Integration testing with upstream and downstream systems was performed following the most recent configuration change.

[SPEC-29] ETHERNET MANAGEMENT SWITCH TRUNKING & VLAN AUDIT

Operational data collected for Ethernet Management Switch Trunking has been aggregated across multiple reporting intervals to establish representative baseline metrics.

Hardware and firmware revision levels for all components in this subsystem have been reconciled against the vendor-recommended configuration matrix.

Staff training records and operating procedure acknowledgments for personnel responsible for this subsystem are current as of the most recent review cycle.

Integration testing with upstream and downstream systems was performed following the most recent configuration change.

[SPEC-30] INFINIBAND SUBNET MANAGEMENT ROUTING PROTOCOLS

Infrastructure parameters associated with InfiniBand Subnet Management Routing Protocols have been validated through a combination of automated monitoring and manual inspection procedures.

Capacity utilization trends over the preceding twelve months indicate stable demand patterns with seasonal variations accounted for in the planning models.

Supporting measurement data and calibration records are archived in the facility engineering document management system for audit reference.

Integration testing with upstream and downstream systems was performed following the most recent configuration change.

[SPEC-31] SWITCH ENCLOSURE POWER CONSUMPTION & THERMAL REJECTION

The technical specifications and measured performance characteristics for Switch Enclosure Power Consumption are documented in the subsections below, with all values reflecting the most recent quarterly assessment period.

Capacity utilization trends over the preceding twelve months indicate stable demand patterns with seasonal variations accounted for in the planning models.

Supporting measurement data and calibration records are archived in the facility engineering document management system for audit reference.

Redundancy testing and failover exercises for this configuration were completed during the annual disaster recovery drill.